

TIMOTHY E. PEER (408)

313-2760

US Citizen

Senior OpenVMS Systems Integrator, and Programmer/Engineer who has delivered software and services to over 150 client sites since 1989. Providing the vision, data-based strategies and cross-technology solutions to integrate critical, often non-interrelated technologies. I possess an extensive skillset with broad coverage of a mix of complex system platforms, configurations, networks topologies, data and factory automation-based systems. Served as a specialist in platform hardening, recovery and unification of a mix of on-prem and cloud-based systems; I communicate effectively with users, technology teams and management.

KEY ACCOMPLISHMENTS

- **OpenVMS (8.4-2L3).** Full stack architecture, Support, Configuration, SDA (all OpenVMS facilities), IA64 (UEFI), Alpha (SRM), VAX – Physical and Virtual – Boot and architecture configuration (emphasis resilient DR and HA - RPO/RTO) to support multiple regions (East/West/Asia) – multi-path CDDI/FDDI – VSI/HP Supported
 - **TCPIP (Full-Stack networking – e.g., DNS, DHCP, MULTINET, TCPIP, TCPWARE)**
 - **DECNET IV/V (migrations and HA)**
 - **LAT/LAST (Terminal Services)**
- **Advanced/Expert Level OpenVMS** (PROMIS/MANMAN) languages (FORTRAN/Forms/DCL/Scripting), Operating System, File Systems, Interconnects, Subsystems (memory, SCA, SDA, Clusters, Networking - DECNET (IV/V), LAT, LAD/TCPIP, RMS, DEC/ORACLE Rdb/ CODASYL DBMS & Oracle RDBMS. Value engagements to improve platform resilience/stability, fault tolerance, high availability.
- **ERP/MES.** MANMAN, Promis MES (5.8/5.9/6.0) – Administration, Programmer of Heterogenous and multisystem VAX/Alpha/IA64/GS (and UEFI – Configuration) clusters - 24x7 high-availability for MES, Manufacturing, Finance, Government, and Defense industries.
- **Oracle Rdb and Codasyl DBMS** – Professional DBA services – Repair, Transaction and Disaster Recovery, OpenVMS Clusters – one client had an inventory of hundreds of Oracle databases. Emphasis on high availability and transaction recovery for off-site DR and cloud hosting. Technology expert designed *Warehouse Suite* -A real-time transaction data replication platform and Framework for Rdb data replication to other SQL92 data sources. Software tools to automate and streamline Oracle Codasyl and Rdb database administration, real-time transaction recovery to DR.
- **Virtualization.** Certification - Stromasys Virtualization Engineer. OpenVMS to Charon/Linux/Windows; Legacy MPE/HP to Virtual Charon HP platform. OpenVMS on single and double virtual stack. Proficiency with VM onprem and off-prem VM Stacks, including Azure and AWS.
- **Multi-Time-Zone Support.** (Various US, UK, EU, Canada, Mexico and Asia) Client engagements.
- **Integration - Dynatrace.** Interface development comprising (> 600 metrics each of 10 systems) with automated updates to SaaS cloud. Integration micro-services were profiled **Dynatrace Perform 2022.**
- **OpenVMS Promis.** Experience “+6 years” System’s Engineer (consultant)/Administrator; PROMIS 5.8.1/5.9, 6.0 – Primary/Secondary/Tertiary on legacy on VAX, Alpha/AXP, Itanium, VSI x86 and Virtual Charon (on Linux and Windows). Supported various 3rd party factory automation tools, EyeLit, Fabtime, Composer, RTI, DMQ, HTTP. Various support included administration, database optimization, user account and static table maintenance.
- **ETL.-** Authored Data Migration tools to/from OpenVMS ERP, SAP, ORACLE, Infor LN, MFG/Pro, QAD.
- **Integration.** OpenVMS RMS, Codasyl DBMS, Rdb to SQL92 sources (DB2, Oracle, Microsoft SQL Server).
- **Data Replication.** Authored an ETL Framework - real-time data from OpenVMS ERP to cross platform SQL DBMS.
- **Micro-Service (Virtual CMS).** Tyler-Odyssey Document Management System – Development on .NET Core 5 – Payment Gateways, Object Model and Micro-Service (API) Development
 - Micro-Service Development - Virtual CMS integration to Tyler/Odyssey
- **Documentation** – System and IT architecture profiling – portable and HTML / Context Searchable.

- **Train-Trainer-Helpdesk.** Training on system, application, surround-strategy integration on topics - hardware, operating system and databases.

PROFESSIONALEXPERIENCE

Skyworks, Sr. Principal Systems Administrator (OpenVMS Systems)

Aug-21 -Present

Reporting to the Director of Global Factory Automation have recommended and implemented high-value OpenVMS initiatives to drive high availability, disaster tolerance and recovery. Proposed/implemented: DECNET IV-V (PWIPDECNET over IP), IPCI (IP as a Cluster Interconnect), Promis HA, OpenVMS Administration, Backup, Recovery, T4 Data Collection; integrated T4 to Dynatrace. The micro-services (Dynatrace integration) development. MES experience includes activities outlined SYSMAN, AUTH and TBL5 table management. The OpenVMS/Promis landscape is considerable with 16+ (Windows/HPUX/OpenVMS) systems serving a multi-locale user community in Korea, Japan, Singapore, various locations throughout North America/Mexico and Canada.

- Upgrades, Extracts, Scripting, User and Table Maintenance
- OpenVMS Disaster Recovery and High Availability Profiling
- Full stack architecture, IA64 (UEFI), Alpha (SRM), VAX – Physical and Virtual – Boot and architecture design to support multiple regions (East/West/Asia) – multipath CDDI/FDDI – VSI/HP.
- Various proposals – architecture, factory integration, decision support, strong emphasis on tech and systems integration
- Functioned as Systems Architect proposed - high availability, system stability, durability and resilient recovery • Provided configuration of OpenVMS interfaces across multiple stacks (hardware, OS and application)
- Authored web based; context search of legacy archived (HP-UX) system data.
- Wrote C# various tools – network payload performance, archival validation, TELNET CSI transformations, text to XML to HTML; developed rules-based AI tools to integrate OpenVMS and Event Management systems, Backup and Recovery – Physical and Virtual systems.

Superior Court - Santa Clara County, Senior Software Engineer

Jun 21-Jul 21

Micro-service development to interface with Tyle/Odyssey CMS, and in-house virtual CMS. Wrote various C# plug-ins DLLs for on-line payments. Provided guidance on the introduction of a common (integrated) object model for use with participating CMS used by the California, Superior Court systems. Submitted proposal to wrap the virtual CMS platform in Microsoft Orleans (Virtual Actors) base system. Resigned due to elder care requirements.

eNVy Systems, San Jose, California / OpenVMS Engineer & Architect

Jul 89 – Jun 21

- Strong emphasis on systems and platform integration. Middleware, tools in C++, C# and VMS Fortran – as necessary to integrate the platform data store to other systems.
- Technical Support – Computer Associates applications, ERP platform, OpenVMS System based applications.
- **Intellectual Property** –
 - Real-Time XML to OpenVMS-Telnet – Scripting Language from .NET/CLR application to Telnet-host based systems.
 - **OpenVMS Outbound FAX** – C# Text-to-PDF transformation design app, service; a language-localized document Windows-based Middleware service - (Lex Tools), real-time conversion of any ASCII document to PDF with SMTP distribution support - e.g., PO, Sales Order, Invoice, Picklist, barcode labels, etc.
 - **Middleware on Windows** – Windows Microservices, a web-based real-time Telnet transactions to an OpenVMS/ERP.
 - A distributed transaction-recovery application – transaction data replication and recovery of hundreds of ORACLE databases to [out of state] disaster recovery systems.
 - Database administration 'ToolKit' application to streamline administration ORACLE databases.

- Significant real-time document transformation and integration to email and fax systems with multi-locale support.
- Authored comprehensive ERP data and platform database management systems integration tools.
- DBMS & Rdb Transaction Recovery – Disaster Recovery Software for ORACLE CODASYL DBMS and Rdb to warm standby DR systems.
- Provided clients with a customized EDI translator (XML2EDI) with a real-time (EDI) interface support to/from OpenVMS ERPs, Global Healthcare Exchange and sales order configurator systems.
- **Client Engagement (Partial List) – for various clients:**
 - Global Healthcare Exchange Interfaces – various EDI interfaces. Real-Time integration to GHX to medical device manufacturers (there were many I integrated/supported).
 - Client industries included aerospace, automotive, healthcare (pharma) and equipment/device, discrete, repetitive and batch process manufacturing, and automotive aftermarket.
 - OpenVMS System Manager/Administrator/Engineer
 - ERP Specialist (MANMAN) Manufacturing, All Modules (MFG, Finance, Logistics, Shop Floor, Planning, Scheduling, Purchasing, Lot Tracing, Service, Process MFG)
 - MES – Factory Automation - PROMIS / Workstream
 - ETL real-time transaction data replication - ERP and real-time data replication - OpenVMS to Microsoft SQL92.
 - File System Repair - engaged (many) database and disk block repair following RAID failures.
 - Disk and Database corruption - repair – Databases include CODASYL DBMS, Oracle Rdb and Oracle RDBMS repair services to file systems on OpenVMS (RMS, Oracle Rdb and Codasyl DBMS page) - client engagements were many.
 - Business Continuation – DR (remediation of catastrophic system & data loss – fire/water.
 - Emphasis on virtualization platform OS, Windows, Linux, HP-UX
 - Architecture - hardware, software and storage and SAN fabric – iSCSI, Windows, OpenVMS, Linux, Hyper-Converged Hardware (SimpliVity), HPE, VMWare, STROMASYS (Alpha/AXP and VAX Virtualization)
 - Hardware (double) Virtualization – Various Platforms - Vax/Alpha-AXP, HP/MPE, HP UX (9000/800) virtualization of AXP (OpenVMS) on Windows, Linux

SELECT TECHNICAL SKILLS

Hardware: VAX, Alpha AXP, IA64, x86, GS-Series, Hyperconverged, OpenVMS Clusters, Linux (RHEL), HP MSA, many other storage systems

ERP: OpenVMS/MANMAN (MG/AP/GL/RP/SY/OM/TK/PJ/PC), ORACLE, PROMIS and Workstream

Language Proficiency: DCL-scripting, C++ (Qt and MFC), C#, PL/SQL, ASP, JSON, REST, MS SQL, Scripts, FORTRAN, .NET, ODBC, Sockets, Xcode (macOS), Precompiled SQL, ORACLE, SQLPRE, SQLMOD

Databases: Oracle Rdb and CODASYL DBMS (Expert), Microsoft SQL Server, DB2, ORACLE, MySQL, SQLite, RMS, ISAM, Git

Operating Systems: OpenVMS (all variants since 5.5), HP and VSI (> 8.4) Supported. Windows, RHEL, macOS, Sun-Unix, HP-UX, Linux

PUBLICATIONS/PRESENTATIONS/WORKSHOP

Have published and presented workshops on: OpenVMS, Tuning, DBA, Oracle Rdb (KODA) Internals, SQL, Data Warehouse, ORACLE, SQL Server, Deriving Data Narratives, ERP

- Symposia – Frequent past presenter & hosted technology workshops on various technology centric topics – storage, performance, database, integration, data delivery, narratives.
- **Integration - OpenVMS T4 / Dynatrace.** OpenVMS Transformation/Ingestion interfaces were profiled, “Dynatrace Perform 2022 in Las Vegas, Nevada - USA”

AUTHORED COPYRIGHTED FRAMEWORK APPLICATIONS

- *RestAPI* – A real-time middleware transaction-centric RESTful web service to Telnet-hosted ERP
- *TelEnvy* – An XML-based scripting language from .NET apps to TELNET based services
- *LexTools* – An XML-template-based text (language-localized) to PDF Conversion Suite of tools.
- *Warehouse Suite* – Real-Time ETL Services From ORA-Rdb and Codasyl DBMS and Microsoft SQL.
 - Qt / MFC / FORTRAN / ODBC32 / SQL
- *Database ToolKit* – Codasyl DBMS and Rdb database administration.

AFFILIATIONS

Board of Directors of CAMUS INTL – OpenVMS Technology Platform Leadership Oracle
Working Group – Oracle Rdb Family (Oracle Codasyl DBMS)

EDUCATION

San Jose State University, De Anza College, Cogswell Polytechnic, Incarnate Word College

Corporate/Professional: ORACLE University, Digital Equipment User's Society, Sun Microsystems, Digital Equipment Corporation, Ask Computer Systems

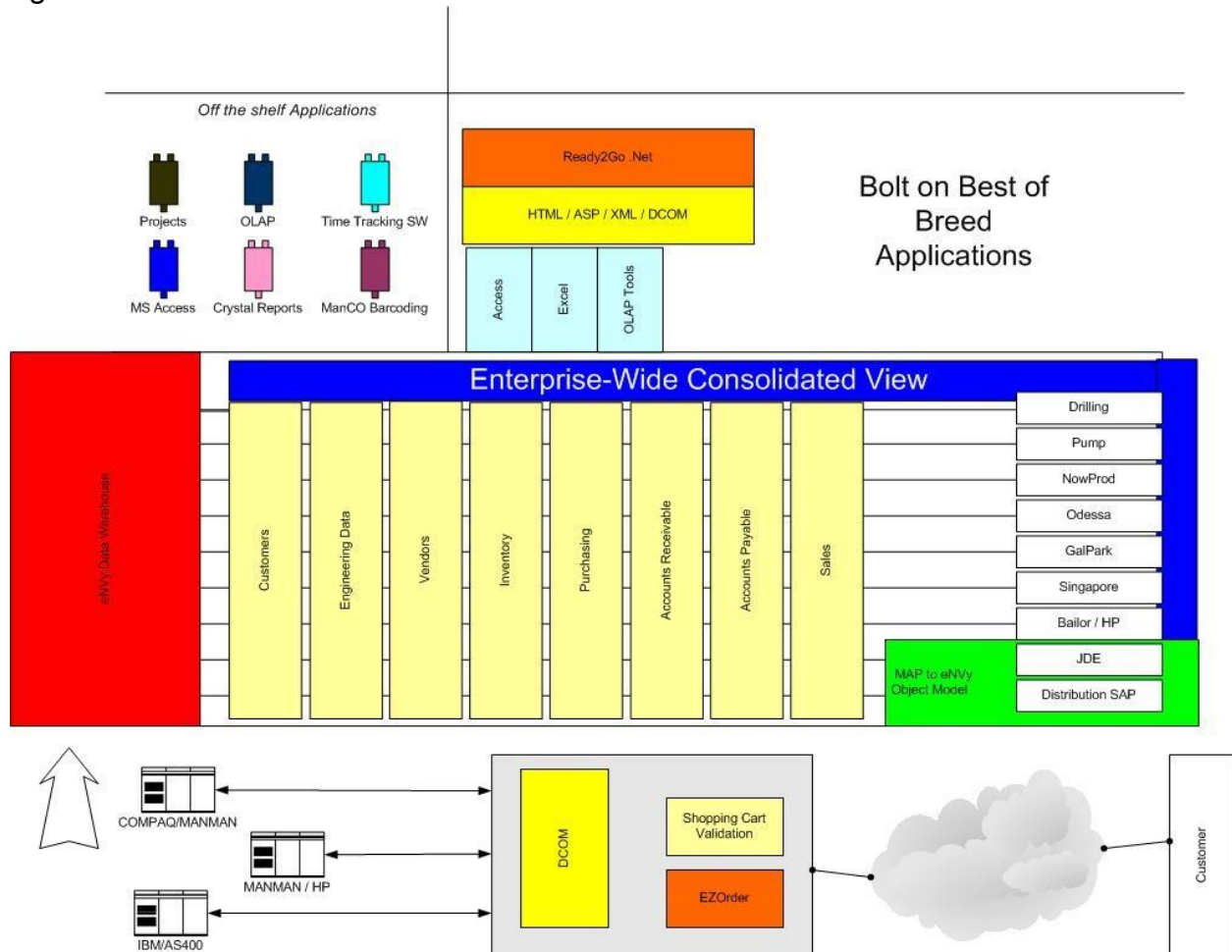
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Have provided numerous proprietary solutions that implemented, migrated, converted and updated ERP host and cloud-based business systems. What follows are client case studies employing proprietary SQL Framework solutions where client specifications were either ambiguous or completely absent. These successful project management examples transcend mere scope of work, project management application, and GANNT/GAP analysis. They evidence a diversity of specialized skills that transformed project/program ambiguity into a manageable matrix of problem, definition, units of work effort, validation, documentation and deployment.

Case Brief #1: Presented a reporting and analytics strawman-project to IT Manager and Engineers for what was then a deficiency of the mainframe ERP. The need was to maintain OLTP integrity and also satisfy the vast requirements of reporting and analytics from ERP data. I provided a proprietary framework, subsequently coined “An Effective Surround Strategy”, for the Mainframe ERP. The framework architecture accommodated multiple organizations (tenants) and bolt-on integration of Windowsbased desktop tools. Subsequently, more than 150 databases have been “surrounded” at multiple client sites across in Europe, North America, Mexico, and Asia.

A key observation is the {red} block to the left which represents the framework infrastructure which permits broad integration across unlimited ERP platforms including SAP. The interface wraps as a “bolt-on” technology to a non-relational OLTP mainframe ERP and presents real-time transaction data from the Mainframe ERP as SQL on Oracle and Microsoft SQL Server to be consumed by stakeholders in all areas of the organization.



This expertise has enabled user-clients to scale and broadly extend the mainframe ERP through proprietary integration interfaces, permitting efficient flow to/from the mainframe ERP to user stakeholders on SQL, and extending the end-life of the ERP by 20 years.

Case Brief #2. The scope of this project was a simple client need, “Provide the ability to execute transactions from .NET applications...” The solution was to extend the “surround strategy” to provide not only advanced analytics and reporting of ERP data, but also to perform real-time transactions from the Windows desktop. After performing a technology (interface) review and providing a “strawman” accompanied by a functional Proof of Concept to the client, a proprietary technology framework with API was built to fully support broad transaction capability from within Windows applications: TelEnvy

Library Framework

(<https://www.envsys.com/wpadmin/customize.php?url=https%3A%2F%2Fwww.envsys.com%2Felementor%2F>).

The proprietary TelEnvy framework provides for dynamic transaction capability using XML (document) transaction units which transmit ERP command sequences to operate the ERP with user-response data. The framework provides “call-back” functionality to receive data from the ERP to the user application for display, process, whatever.

Case Brief #3. Extended the TelEnvy Framework to send/receive data from a hosted AWS/EC2 portal to the Mainframe ERP. The client needed to communicate in real-time with the ERP from AWS/EC2. The client asked me to provide middleware within a sixmonth project to create a real-time bridge between the AWS/EC2 web service and the mainframe ERP.

For this project, there was no specification, documentation, scope of work or technology review, and a mildly cooperative third- party web-service developer group located in Bangalore India. On the mainframe side was an ERP consultant-programmer. To build an interface required the data clues be inferred from minimal data input/output data elements enumerated in Microsoft Excel worksheet documents. What seemed apparent was the eventual necessity of the middleware to support the receipt and handoff of data for any number of transaction-sets (derived Excel worksheet documents). Thus, the project scope evolved from a fixed middleware interface between AWS and the ERP to a framework which supported not just AWS but any provider/consumer service on both sides of the middleware interface.

The approach leveraged the technologies which I previously developed and incorporated available framework technologies via existing Open-Source libraries. After a considerable search of the Open-Source libraries, I employed data envelopes which had the broadest interface options for SAP and other data stores. Candidate data envelopes from/to AWS support data envelopes created in: SOAP, REST, EDIFACT, FTP, SFTP and flat file. The transaction engine provides for data cleansing and transformation between transaction interfaces in real-time. As such, the middleware deployed was highly agile/adaptable and responsive. The framework was evaluated for transaction processing from AWS/EC2 to systems in Romania on IBM/AS400, and ERPs executing in Central America.

Subsequent performance testing revealed the framework could handle any number of simultaneous transaction requests to most ERP host-based systems, and for this implementation achieved sub-second response times (e.g. process a sales order with 99 lines – completed in < 7 seconds) for minimal real-time transaction delays.

Case Brief #4. A client who was running a mainframe-based FAX application was notified that the FAX product would no longer be supported. The client had been FAXing

purchase order documents to suppliers with the embedded “PNG-signature” of a procurement manager who had left the company long ago. The client had a single requirement: replace the deprecated mainframe FAX with an updated service: “update/replace the PNG-signature”, and do so at minimal cost.

I reviewed the then-commercially available applications for a comparable replacement solution. Available solutions were very high cost and beyond the client’s budget. The client asked that we provide an alternate, more cost-effective solution. As such, I wrote a suite of ASCII file-to-PDF tools in Microsoft .NET (C#): “Lex-Suite”. The proprietary Lex Suite toolset includes a forms designer application (similar to a Crystal Reports designer), a Windows Services application and a Mainframe-based printer symbiont to redirect printed output from the Mainframe to a Lex Suite Windows Server.

The resulting project yielded a scalable solution which supported internationalization/localization for five languages, and more than 30 forms (including purchase orders). The solution scaled readily to add new languages and forms easily and quickly (in 20 minutes or less).

Case Brief #5. While employed at a Semiconductor Equipment company as an IT Manager, analyzed, demonstrated to all levels (include BOD) a telephony and network reengineering project to support an ERP (Finance, AP, AR, Manufacturing, Field Services, G/L, Projects – MRPII – many tenant subsidiary companies).

Prepared RFQ - with coordination of the corporate purchasing group to: AT&T, Sprint, INFONET and MCI.

This project consisted of many dimensions – technology, finance, corporate “politics”, legal & tariff review, validation, cost of service, cost avoidance... The project was championed by the VP of Field Service in an attempt to reduce CS department allocations to his department for packet switched (x.25) service from corporate to remote offices. It was then subsequently expanded in scope to include all telephony, networking and video conferencing. The project revealed a rather dysfunctional infrastructure and a web of contracts and stakeholders, the results of which measured at > .80/minute to make a phone call. The project required significant user-coordination and an elaborate feedback mechanism to reign-in all interested parties to adopt an updated telephony and (adopt a corporate) network platforms. The project was completed in under a year, the result became an immediate application platform to implement a video conferencing center and reduce costs of travel. The project consisted of many dimensions, and stakeholders were not always cooperative and quite skeptical of changing to a newer telephony and networking technology. The project was estimated to provide an additional \$.03-05 EPS. The actual EPS additional value was estimated to be much higher given the advanced network and telephony infrastructure.